

Cover letter

Dear Editors,

I am writing to submit our manuscript titled "**Causal discovery in urban data with temporal empirical dynamic modeling**" for consideration for publication in the *Computers, Environment and Urban Systems*.

This study introduces *tEDM*, an open-source R package that advances empirical dynamic modeling for temporal causal discovery in complex urban systems. By adapting and integrating core EDM-based methods, *tEDM* provides a framework that accommodates heterogeneous, high-frequency, multivariate, and spatially replicated urban time series. Applications to case studies on air pollution and cardiovascular health in Hong Kong, county-level carbon emissions and temperature in the United States, and the spread of COVID-19 across Japanese prefectures demonstrate its capacity to uncover nonlinear interactions, feedback mechanisms, and evolving causal linkages. The package offers both methodological contributions and practical analytical tools, thereby enhancing the capacity of urban science to achieve deeper causal understanding and to support evidence-based decision-making.

Author contributions: Wenbo Lv, Yongze Song and Wufan Zhao conceived the idea and designed the study; Wenbo Lv, Yangyang Lei, Yongze Song, Shaoqing Dai and Wufan Zhao collaborated on the implementation of the *tEDM* package in this manuscript; Yangyang Lei, Wen Yi, Xiao Li, and Yiming Qin collected and analyzed the case studies; Wenbo Lv and Yangyang Lei wrote the first draft, and all authors contributed to the writing and editing of the final manuscript.

The authors declare no competing interests.

Please do not hesitate to contact me should you require additional information. We appreciate your kind consideration.

Best regards,

Sincerely,

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